Physics-Informed Neural Networks for the One-Dimensional Convection–Diffusion Equation

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November 13, 2026

Abstract

We compare physics-informed neural networks (PINNs) with a limited Lax–Wendroff finite-volume scheme for the one-dimensional linear and nonlinear convection–diffusion equation with a discontinuous initial condition, an oscillatory Dirichlet inflow, and a zero-gradient outflow. A three-grid Richardson extrapolation of the limited Lax–Wendroff scheme provides the reference. Its L^1 self-convergence estimate is 1.3–5.6×10^{-5} for $\nu \geq 0.002$ and 1.4–3.9×10^{-4} at the stress-test value $\nu = 10^{-4}$; the corresponding L^∞ estimate on the common evaluation grid is 5.7×10^{-4}–5.4×10^{-3} and 4×10^{-2}–10^{-1}, respectively. Across six cases, the limited scheme has four to nine times lower L^1 error than a baseline tanh PINN at $\Delta x = \pi/200$. The PINN's L^∞ error is dominated by the initial datum: a hard branch that imposes the datum exactly removes this contribution without adding parameters. Across the exact-IC sweep of eight plain architectures (five seeds each), the mean L^∞ is 0.25–0.43 on the linear problem and 0.18–0.35 on the nonlinear problem, compared with 0.70–0.80 for the soft-IC 4×64 baseline. The lowest mean PINN values (0.25–0.27 linear, 0.18–0.19 nonlinear) remain about 2.5–3 times the finite-volume values on the same grid. Exact initial conditions therefore close neither the L^1 nor the L^∞ gap. The lowest observed mean linear L^1 is the 4×64 sine network; larger or deeper sine networks do not improve the tradeoff. The lowest observed mean nonlinear L^∞ is the 4×64 or 6×64 tanh network; differences among the best architectures are within one standard deviation in both cases, so these rankings are indicative rather than statistically resolved. All PINN runs use 3000 epochs; a baseline run takes about 2.4 min on a single NVIDIA GeForce RTX 3060, and the hard-IC sweep was run with multiple jobs concurrently on the same GPU.

1 Background: the convection–diffusion equation

The convection–diffusion equation couples advection, which transports u with speed $f'(u)$, with diffusion, which smooths gradients at rate ν. For $f = au$ a step becomes a heat front of thickness $O(\sqrt{\nu t})$; for $f = u^2/2$ (Burgers' equation) a compressive step steepens into a shock of width $O(\nu/\Delta u)$, whereas the expansive step used here opens into a rarefaction, and the oscillatory inflow generates shocks downstream. Thin fronts are hard for low-order finite-volume schemes unless limited [3], and for smooth tanh networks, which are spectrally biased toward low frequencies [5] and need explicit high-frequency features to fit a moving discontinuity [8]. The convection-dominated limit is difficult for PINNs. We ask how a baseline PINN compares with a limited finite-volume scheme on this problem, whether exact initial conditions close the L^∞ gap, and which architecture is best once the initial layer is removed.

2 Problem formulation

We solve

$$u_t + f_x = \nu u_{xx}, \qquad 0 \le x \le \pi, \quad t \ge 0, \qquad f = \begin{cases} u & \text{linear,} \\ u^2/2 & \text{nonlinear,} \end{cases} \tag{1}$$

with the discontinuous initial condition

$$u(x,0) = \begin{cases} 1/2 & x < \pi/2, \\ 3/2 & x \ge \pi/2, \end{cases} \tag{2}$$

the oscillatory Dirichlet inflow condition

$$u(0,t) = 1 - \tfrac{1}{4}\cos(4t) - \tfrac{1}{4}\cos(8t), \tag{3}$$

and the zero-gradient outflow condition $u_x(\pi,t) = 0$. We consider $\nu \in \{0.012, 0.004, 0.002\}$ and, as a stress test, $\nu = 10^{-4}$. For $\nu = 0$ the linear problem has the closed-form traveling solution used to validate the classical solvers.

With unit advection speed the final time $t = \pi$ is one flow-through. The initial jump leaves the domain before $t = \pi$, but its diffusive imprint persists and remains difficult for the network to represent. The inflow supplies a continuously varying signal at frequencies $\omega \in \{4, 8\}$, and the solution contains a smooth traveling pulse together with a sharp, decaying front.

3 Methods

3.1 Classical finite-volume scheme and reference

The comparison solver combines a limited Lax–Wendroff convective flux with a trapezoidal (Crank–Nicolson) diffusion term and solves a tridiagonal system for the interior unknowns. It uses the van Leer monotonized-central (MC) limiter. The left boundary is Dirichlet; the right boundary uses the zero-gradient condition $u_{j_{\max}} = u_{j_{\max}-1}$. All finite-volume and reference solves use the CFL number $\Delta t/\Delta x = 0.5$.

The limiter is applied to the convective interface flux in the standard limited Lax–Wendroff form [3]; the diffusion terms are centered in time and space.

The reference is the three-grid Richardson extrapolation of the limited Lax–Wendroff (van Leer MC) scheme. Because the limited scheme is only first order across the moving discontinuity, a two-grid combination with $p = 2$ leaves the leading $O(\Delta x)$ front term. We combine $\Delta x = \pi/400$, $\pi/800$ and $\pi/1600$,

$$u_{\text{ref}} = u_1 + 2\,(u_1 - u_2) - \tfrac{1}{3}\,(u_1 - u_4), \tag{4}$$

with u_1 the $\pi/1600$ solution and u_2, u_4 the $\pi/800$ and $\pi/400$ fields interpolated onto the fine grid. This cancels both the $O(\Delta x)$ and $O(\Delta x^2)$ terms in an averaged sense; the cancellation is not pointwise at the front. The combination is applied only for $t > 0$; at $t = 0$ the reference is set to the exact initial datum, because extrapolating across the discontinuity produces nonphysical pointwise values.

Repeating the combination one grid finer changes the field by an L^1 of 1.3–1.6 $\times\, 10^{-5}$ at $\nu = 0.012$, 1.4–3.0 $\times\, 10^{-5}$ at $\nu = 0.004$, and up to 5.6 $\times\, 10^{-5}$ at $\nu = 0.002$; at $\nu = 10^{-4}$ it is 1.4–3.9 $\times\, 10^{-4}$. On the common 201×101 evaluation grid the corresponding L^∞ changes are 5.7 $\times\, 10^{-4}$–1.5 $\times\, 10^{-3}$ for $\nu \ge 0.004$, up to 5.4 $\times\, 10^{-3}$ at $\nu = 0.002$, and 3.9 $\times\, 10^{-2}$–1.0 $\times\, 10^{-1}$ at $\nu = 10^{-4}$. The L^∞ error is localized at the front and is much larger than the L^1 error. We take the repeated-combination differences as the self-convergence estimate.

3.2 Physics-informed neural network

The surrogate is a physics-informed neural network [6]: a fully connected feed-forward network $u_\theta : \mathbb{R}^2 \to \mathbb{R}$ that takes the raw space-time coordinates (t, x) as a two-dimensional input and returns the solution at that point. The baseline architecture is an input projection to width $H = 64$, four hidden-to-hidden layers of width 64, and a linear output, with tanh activations and 16,897 trainable parameters; we label this network 4×64. The input is not rescaled; the network sees the physical coordinates in $[0, \pi]$, so no coordinate features are supplied.

The parameters are found by minimizing the composite loss

$$\mathcal{L} = \lambda_{\text{pde}}\mathcal{L}_{\text{pde}} + \lambda_{\text{ic}}\mathcal{L}_{\text{ic}} + \lambda_{\text{bc}}\mathcal{L}_{\text{bc}}, \tag{5}$$

where $\mathcal{L}_{\text{pde}}$ is the mean-squared PDE residual $u_t + f_x - \nu u_{xx}$ evaluated by automatic differentiation at interior collocation points, and $\mathcal{L}_{\text{ic}}$, $\mathcal{L}_{\text{bc}}$ are mean-squared penalties enforcing the initial and boundary data (including the zero-gradient outflow condition). The residual is evaluated on $N_{\text{pde}} = 3000$ uniformly sampled interior points, with $N_{\text{ic}} = 128$ initial samples and $N_{\text{bc}} = 128$ samples per boundary. Each epoch shuffles the collocation points and takes mini-batches of size 256. We use Adam with learning rate 10^{-3}, decayed by 0.5 at epochs 1200 and 2200, and fixed weights $(\lambda_{\text{pde}}, \lambda_{\text{ic}}, \lambda_{\text{bc}}) = (1, 10, 1)$. All PINN results below are trained for 3000 epochs on a single NVIDIA GeForce RTX 3060 (12 GB). The hard-IC sweep was executed with multiple jobs concurrently on the same GPU, so its per-run wall-clock times reflect GPU contention rather than the cost of the exact-IC branch, which adds negligible work.

3.3 Architecture variants

We consider two activations, tanh and sine, at four sizes: 2×32, 4×64 (baseline), 6×64, and 4×128, spanning 2,241 to 66,561 parameters. The sine networks use $\sin(3z)$ and the SIREN initialization of Sitzmann et al. [7]. This gives eight architectures; each is trained from five seeds on the linear and nonlinear problems at $\nu = 0.012$ under the common 3000-epoch budget.

3.4 Exact initial condition

All architectures in Section 3.3 use the same treatment of the discontinuous datum. Let u_θ denote the raw network output. The model used for training and evaluation is

$$\tilde{u}_\theta(t, x) = \begin{cases} u_0(x), & t = 0, \\ u_\theta(t, x), & t > 0, \end{cases} \tag{6}$$

where u_0 is the step of Eq. (2). At the $t = 0$ collocation and evaluation points the output is the exact datum, independent of the network parameters; the branch adds no parameters. The PDE and boundary terms of Eq. (5) are evaluated on $\tilde{u}_\theta$; the IC term is evaluated on the raw field u_θ, because $\tilde{u}_\theta$ is identically u_0 at $t = 0$ and would give no gradient. The penalty keeps the $t > 0$ branch close to the datum as $t \downarrow 0$, so the branch does not introduce a large inconsistency.

This is a hard-constraint construction in the spirit of [1, 2, 4], applied here to a discontinuous datum. Rather than the usual $u_0(x) + tN_\theta(x, t)$ ansatz, which suppresses the network correction near $t = 0$, we switch branches at $t = 0$ and train the $t > 0$ branch with the IC penalty. The soft-constraint baseline uses the same architectures without the branch and with the IC penalty applied directly to the network output.

4 Results

Errors are reported as L^1 (mean absolute) and L^∞ (maximum) on a common 201×101 space-time grid against this reference; the finite-volume fields are sampled at the same time levels as the PINN. Because the maximum is sensitive to the time sampling, all methods are scored on this common grid. The finite-volume scheme starts from the exact discontinuous datum, so its error vanishes at $t = 0$, whereas the soft-IC PINN must learn the initial condition and its largest error is there. These norms are used only for evaluation; training minimizes the composite loss of Eq. (5).

4.1 Accuracy over the six cases

The finite-volume scheme is consistently more accurate than the baseline PINN. At $\Delta x = \pi/200$ its L^1 error on the common grid is $(1.6\text{–}2.1) \times 10^{-3}$, compared with $(0.8\text{–}1.6) \times 10^{-2}$ for the PINN, so the scheme has four to nine times lower error; the largest ratio is 9.4 (linear, $\nu = 0.004$) and the smallest 4.2 (nonlinear, $\nu = 0.012$). The gap is similar for both problems. PINN values are five-seed means at the 3000-epoch budget, scored against the same reference as the finite-volume solutions.

The soft-IC baseline's L^∞ error is 0.70–0.80. Most of it is at $t = 0$: removing that frame lowers the seed-averaged error to 0.22–0.47, still above the finite-volume 0.07–0.20 on the same grid. Some individual runs remain as high as 0.54 even after removing that frame, so the initial layer explains the average but not every run. The L^1 error behaves differently: removing $t = 0$ leaves it nearly unchanged, consistent with the initial layer being a small fraction of the space-time average; the exact-IC results below confirm that exact initial conditions do not close the L^1 gap. With soft constraints, architecture changes help only unevenly; a sine activation brings the linear $\nu = 0.012$ error to 2.3 times the finite-volume value but does not improve the nonlinear problems. Section 4.4 shows that exact initial conditions reduce the L^∞ gap, but the finite-volume scheme remains more accurate.

4.2 Linear convection–diffusion

Figure 1 shows the final-time profile of the exact-IC sine network; at this time it tracks the reference closely, with the visible deviation near the outflow. The limited scheme is second order in smooth regions but only first order across the moving front, so whole-grid L^1 convergence orders are about 1.0–1.4. At $t = 0$ the soft-IC PINN's L^∞ error is at $x = \pi/2$. An optimally smeared step would give a half-jump error of 0.5; here the error is larger because the learned transition is displaced as well as smoothed: at $x = \pi/2$ the network returns 0.69 where the Richardson-extrapolated solution is 1.5, an error of 0.81. As the solution evolves, the maximum error over the domain decays (from 0.81 at $t = 0$ to 0.27 at $t = 0.5$ and 0.08 at $t = 2$), consistent with an error that originates at the initial datum. The exact initial condition of Section 3.4 removes this contribution; the residual whole-grid L^∞ occurs shortly after $t = 0$ at the front, while at the final time the profile is nearly indistinguishable from the reference except near the outflow.

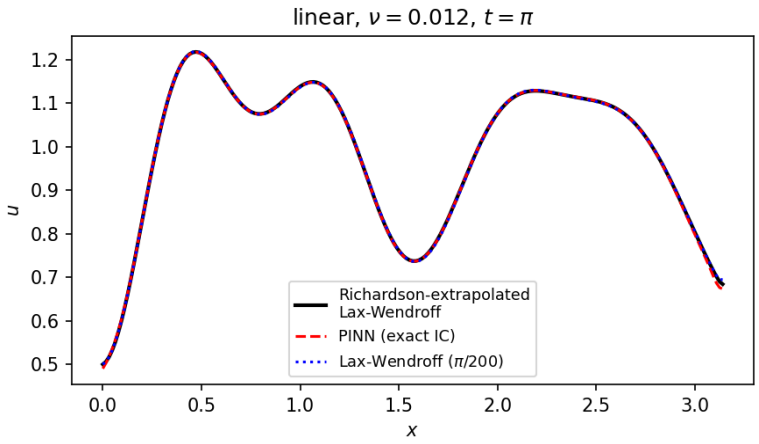


Figure 1: Linear $\nu = 0.012$: final-time profiles for the exact-IC 4×64 sine network (seed 2). The solid black curve is the Richardson-extrapolated Lax–Wendroff reference, the red dashed curve is the PINN, and the dotted blue curve is the limited Lax–Wendroff solution at $\Delta x = \pi/200$.

4.3 Low-diffusivity stress test

As ν decreases the front sharpens. At $\nu = 10^{-4}$ the exact-IC 4×64 sine model (linear) and the 4×64 tanh model (nonlinear), each with three seeds, have $L^\infty = 0.68 \pm 0.19$ and 0.63 ± 0.05, compared with 0.34–0.41 for the limited scheme on the same grid; their L^1 errors are 8.3×10^{-3} and 4.1×10^{-2}, versus 2.4×10^{-3} and 3.9×10^{-3} (Figure 2). The exact IC removes the initial-layer contribution (the soft-IC baseline had $L^\infty \approx 0.87$–0.90), but the moving front and, in the nonlinear case, the downstream shocks now dominate; in the linear case the largest errors for two of the three seeds are at the outflow.

The reference L^∞ self-convergence on the common grid at this value is 3.9×10^{-2} (linear) and 1.0×10^{-1} (nonlinear), so the stress-test L^∞ values carry a reference uncertainty of roughly 10–25% for the finite-volume scheme and 6–16% for the PINN. The ordering is unchanged, but the absolute values are less precise than at $\nu = 0.012$. The classical L^1 continues to decrease under refinement (to 9.0×10^{-4} and 1.9×10^{-3} at $\pi/400$). The nonlinear L^∞ at the unresolved viscous shock stays near 0.35–0.41, as expected while the shock remains under-resolved; the linear L^∞ falls to 0.19 at $\pi/400$ as the diffusive front is resolved.

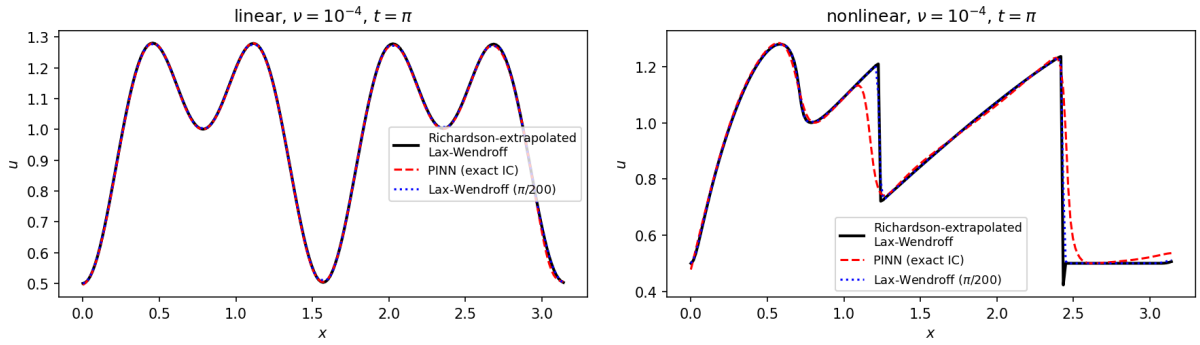


Figure 2: Stress test at $\nu = 10^{-4}$: linear (left) and nonlinear (right). The solid black curve is the Richardson-extrapolated Lax–Wendroff reference, the dotted blue curve is the limited Lax–Wendroff solution at $\Delta x = \pi/200$, and the red dashed curve is the exact-IC PINN (4×64 sine for linear, 4×64 tanh for nonlinear).

4.4 Exact initial condition and architecture

Table 1 compares the eight plain architectures of Section 3.3 with exact initial conditions, each trained from five seeds at $\nu = 0.012$. The branch adds no parameters, so the exact-IC rows are architecture-for-architecture comparable. The table includes soft-IC controls only for the two 4×64 networks; for those, exact IC removes the initial-layer error, and across the full exact-IC sweep the mean L^∞ is 0.25–0.43 on the linear problem and 0.18–0.35 on the nonlinear problem, compared with 0.70–0.80 for the soft-IC 4×64 baseline. The lowest means remain above the finite-volume values (0.09 linear, 0.07 nonlinear): on the nonlinear problem the 4×64 and 6×64 tanh networks give the lowest L^∞, while the 4×64 sine network gives the lowest L^1. On the linear problem the 4×128 sine network gives the lowest L^∞ and the 4×64 sine network the lowest L^1. Differences among the best few architectures are within one standard deviation, so these rankings are indicative rather than statistically resolved.

Larger or deeper sine networks do not improve the linear tradeoff. The 6×64 sine network becomes unstable on the nonlinear problem, where one seed reaches $L^1 = 0.207$, and the 4×128 sine network gives a higher L^1 than the 4×64 sine network on both problems. The 4×64 sine network is therefore the better sine choice for L^1, while the 4×128 sine network buys only a small linear L^∞ gain.

Exact IC is primarily an L^∞ correction and closes neither gap: the best L^1 and L^∞ values remain about 2.5–3 times those of the finite-volume scheme. On the linear problem it slightly worsens L^1 relative to the soft-IC sine baseline, while on the nonlinear problem it improves it; both differences are within one standard deviation. Once the initial layer is removed, architecture choice sets the remaining accuracy, but the finite-volume scheme remains the more accurate method.

5 Discussion and conclusion

For this problem the limited finite-volume scheme remains the more accurate method: about 2×10^{-3} L^1 at $\Delta x = \pi/200$, against 10^{-2} for the baseline PINN. The PINN's L^∞ error, however, is dominated by the discontinuous initial datum rather than by the interior solution, and imposing the datum exactly removes that contribution at no parameter cost. In the architecture sweep, exact IC reduces the mean L^∞ to 0.18–0.43, but the finite-volume scheme remains about 2.5–3 times more accurate in both norms. It does not close either gap, and at $\nu = 10^{-4}$ the moving front and the

Variant	Setting	Linear		Nonlinear	
		L^1 $[10^{-3}]$	L^∞	L^1 $[10^{-3}]$	L^∞
Soft IC (5 seeds, 3000 epochs)					
Baseline	tanh, 4×64	13.7±5.4	0.77±0.23	8.41±5.41	0.73±0.20
Sine	sin, 4×64	4.46±1.88	0.74±0.19	8.85±4.27	0.73±0.18
Exact IC (5 seeds, 3000 epochs)					
Tanh	2×32	20.6±6.1	0.34±0.11	20.4±7.7	0.30±0.11
Tanh	4×64	13.1±4.4	0.35±0.20	6.40±2.9	0.19±0.11
Tanh	6×64	11.4±5.9	0.43±0.20	6.98±1.7	0.18±0.08
Tanh	4×128	8.75±4.1	0.33±0.15	8.62±5.1	0.28±0.16
Sine	2×32	5.96±1.7	0.27±0.10	10.4±4.1	0.23±0.10
Sine	4×64	4.85±2.1	0.27±0.15	5.80±1.6	0.21±0.10
Sine	6×64	12.1±7.8	0.27±0.11	54±77	0.35±0.23
Sine	4×128	9.62±5.1	0.25±0.11	18.0±6.6	0.28±0.09
Finite volume					
Lax–Wendroff (limited), $\pi/200$		1.92	0.09	2.01	0.07

Table 1: PINN and finite-volume comparison at $\nu = 0.012$. PINN entries are mean±s.d. over five seeds; predictions are not ensembled. All errors are evaluated on the common 201×101 space-time grid. The soft-IC rows use the two 4×64 architectures without the exact-IC branch; the exact-IC rows use Eq. (6). The Lax–Wendroff row is a single deterministic solve.

outflow dominate again; the reference L^∞ uncertainty at that value (4×10^{-2}–10^{-1}) is also larger than at $\nu = 0.012$.

The lowest mean architecture depends on the target: the 4×64 sine network for linear L^1, 4×128 sine for linear L^∞, and the 4×64 or 6×64 tanh network for nonlinear L^∞; these differences are within seed variability. None of these choices changes the overall conclusion. For this problem the limited scheme remains the better choice. For a PINN, the exact initial condition is the most effective change for L^∞ among those tested, whereas the sine activation gives the largest linear L^1 improvement; further gains require resolving the moving front rather than the initial datum.

Figure 3 shows final-time profiles for both problems. The exact-IC PINN tracks the reference and the limited scheme over most of the domain; the visible differences are at the moving front and near the outflow, where the residual L^∞ error remains.

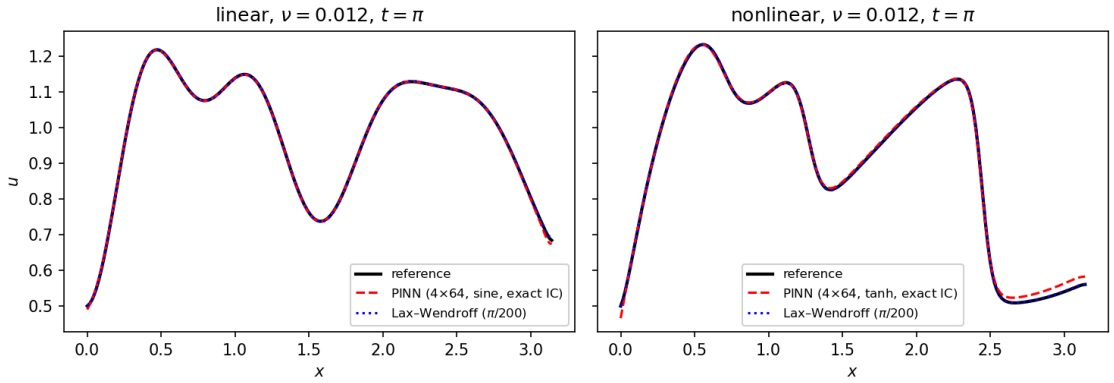


Figure 3: Final-time profiles at $\nu = 0.012$. Linear (left): exact-IC 4×64 sine network (seed 2). Nonlinear (right): exact-IC 4×64 tanh network (seed 0). Solid black: reference; red dashed: PINN; dotted blue: limited Lax–Wendroff at $\Delta x = \pi/200$. The PINN overlaps the reference except near the moving front and the outflow.

Acknowledgment

The code and text were developed with assistance from the DeepSeek V4.1 Flash language model.

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